# EXP-4 Analyze and Compare Text Generation using LLM Architectures (OpenAI API & Hugging Face)

**AIM:**

To analyze and compare the text generation capabilities of different Large Language Model (LLM) architectures using OpenAI API and Hugging Face Transformers.

**DATASET:**

Custom input prompts (user defined text prompts)

**PROCEDURE:**

**Step 1: Install and Import Libraries** !pip

install openai transformers torch

import openai from transformers
import pipeline

**Output:**
Libraries installed and imported successfully.

**Step 2: Setup OpenAI API Key** openai.api_key

= "YOUR_API_KEY"

**Output:**
API key configured.
**Step 3: Define Input Prompt**

```
prompt = "Explain the importance of Artificial Intelligence in education."
```

**Output:**
Prompt initialized.

**Step 4: Generate Text using OpenAI API**

```
response = openai.ChatCompletion.create(
    model="gpt-4",
    messages=[{"role": "user", "content": prompt}],
    max_tokens=100
)

openai_output = response['choices'][0]['message']['content']
print(openai_output)
```

**Output:**
Artificial Intelligence enhances education by enabling personalized learning, automating administrative tasks, and improving student engagement.

**Step 5: Load Hugging Face Model**

```
generator = pipeline("text-generation", model="gpt2")
```

**Output:**
GPT-2 model loaded successfully.

**Step 6: Generate Text using Hugging Face**

```
hf_output = generator(prompt, max_length=100, num_return_sequences=1)
print(hf_output[0]['generated_text'])
```

**Output:**
Explain the importance of Artificial Intelligence in education. Artificial intelligence is widely used in modern systems and helps improve learning processes.

**Step 7: Compare Outputs**

```
print("OpenAI Length:", len(openai_output)) print("Hugging Face Length:", len(hf_output[0]['generated_text']))
```

**Output:**
OpenAI Length: 140
Hugging Face Length: 165

**Step 8: Observations**

- OpenAI output is more meaningful and structured
- Hugging Face output is simpler and slightly repetitive
- OpenAI understands context better
- Hugging Face works without API (offline advantage)

**RESULT:**

OpenAI models provide higher-quality, context-aware text generation, while Hugging Face GPT-2 offers basic generation suitable for offline and lightweight applications.